

Samuel James Hale

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Education

East Carolina University — M.S. in Computer Science

Jan. 2023 – May 2026

GPA: 4.0

East Carolina University — B.S. in Software Engineering, Concentration in Data Science

Aug. 2021 – May 2025

Chancellor's List (Spring 2025), Dean's List (Fall 2022, Fall 2023) · BS/MS Accelerated Program

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C#, C/C++, SQL, Go, Bash
Web & Frontend: React, Angular, Express.js, HTML/CSS, RESTful API Design, GraphQL, HTMX
Backend: Spring Boot, Django, Flask, Micronaut, Node.js, .NET, Microservices, gRPC
Databases: PostgreSQL, MySQL, SQL Server, SQLite, Neo4j, Redis; query optimization, indexing, RBAC
Cloud & DevOps: AWS (Lambda, RDS, IoT Core, Cognito, EC2, S3), Azure, Docker, Kubernetes, Terraform, Linux, CI/CD (GitHub Actions, Jenkins, GitLab CI/CD, Azure DevOps)
Testing & Quality: JUnit, pytest, unit/integration testing, test-driven development (TDD), code review
Security: OWASP Top 10, JWT/OAuth 2.0 authentication, secure coding practices, input validation
AI & Tooling: Claude Code, GitHub Copilot, LLM-assisted development, prompt engineering
Version Control: Git, GitHub, GitLab; pull request workflows, branching strategies, SVN
Process: Agile/Scrum, sprint planning, system design, technical documentation

Relevant Experience & Projects

Personal Project — Ultimate SWE | (AI Agent Tooling, Prompt Engineering, Technical Writing)

2025

github.com/SamuelH98/Ultimate-SWE

- Designed and published a structured workflow skill that instructs AI coding agents (like Claude Code) how to approach software projects the right way — covering requirements, architecture, testing, and handoff in a defined order rather than jumping straight to writing code.
- Wrote individual workflow files for each phase of development that the agent follows as instructions, making its behavior more predictable, thorough, and consistent across different projects.
- Built it around the idea that AI agents produce better software when they're given the same discipline a good engineer would use — understand the problem first, design before coding, test against real requirements, and document what was built.

Full-Stack Developer — Senior Capstone | (Django, PostgreSQL, Python, GitLab CI/CD)

Aug. 2024 – May 2025

NC Math Placement Web Application · Team of 12

- Part of a 12-person student team that built a real web application used to place North Carolina students into the right math courses — I worked on both the front end (what users see) and the back end (the logic and database behind the scenes).
- Created a feature that automatically generates formatted documents from the app's data, saving the team from doing it by hand every time and making sure the output looked consistent no matter who ran it.
- Fixed several visual bugs in the app's interface that were confusing users, then worked with teammates to get those fixes added to the shared codebase without causing new problems.
- Helped keep our automated test pipeline running smoothly so that every time someone submitted new code, the system checked it for problems automatically — this meant we caught mistakes early instead of finding them after launch.